Edmonds–Karp Maximum Flow Visualization Using Breadth-First Search (BFS)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by

Y. Uttej (2520030439)
G. Hitesh (2520030370)
K. Sudheer (2520030470)

Team Number: 23

Section: 4

Under the guidance of

J. Yamini Devi

Faculty Guide, Department of Computer Science and Engineering

Hyderabad-500090, Telangana, India.

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



DECLARATION

The Mini Project Report entitled "Edmonds–Karp Maximum Flow Visualization Using Breadth-First Search (BFS)" is a record of Bonafide work of Y. Uttej (2520030439), G. Hitesh (2520030370), K. Sudheer (2520030470) , submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering/Computer Science and Information Technology/ Artifical Intelligence & data Science to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

Y. Uttej (2520030439)

G. Hitesh (2520030370)

K. Sudheer (2520030470)

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the mini project based report entitled “Edmonds–Karp Maximum Flow Visualization Using Breadth-First Search (BFS)” is a bonafide work done and submitted by Y. Uttej (2520030439), G. Hitesh (2520030370), K. Sudheer (2520030470), in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in Department of Computer Science Engineering/ Computer Science and Information Technology/Artificial Intelligence & Data Science, K L (Deemed to be University), during the academic year 2026-2027.

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my guide, <<Guide Name>>, <<Designation>>, <Department>> which cannot be expressed in words for his/her tremendous support, encouragement and guidance for this project.

We express our gratitude to Dr. P.Venkateshwar Rao (CSE)/Dr. Khaja Shareef (CSA)/ Dr. Sreenath Itikela (AI&DS), Head of the Department for Computer Science Engineering/ Department for Computer Science and Information Technology/Department of Artificial Intelligence and Data Science for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

Y. Uttej (2520030439)

G. Hitesh (2520030370)

K. Sudheer (2520030470)

TABLE OF CONTENTS

S.No.	Contents	Page no.
1.	Abstract	
2.	Introduction	
3.	Literature survey	
4.	Problem Statement and Objectives	
5.	Hardware and Software requirements	
6.	Implementation	
7.	Results & Discussion	
8.	Conclusion & Future Scope	
9.	References	

1.ABSTRACT

The Edmonds–Karp algorithm calculates maximum flow in a directed network by using Breadth-First Search (BFS) to find augmenting paths in a residual graph. This project visualizes the network, BFS exploration, augmenting paths, residual capacities, and flow updates step by step. The algorithm repeats the process until no augmenting path remains; the resulting flow is the maximum flow from source to sink. The visualization helps learners understand graph representation, queues, BFS traversal, residual edges, and flow augmentation.

Team Name: Edmonds–Karp

Team Logo (if any):

Team Members:

1.
2.
3.
4.
5.

2. INTRODUCTION

Maximum flow is a graph problem that finds the greatest possible flow from a source vertex to a sink vertex without exceeding edge capacities. Edmonds–Karp solves this problem by applying BFS to select shortest augmenting paths in the residual graph. This project presents the algorithm as a step-by-step visualization.

The report presents related work, the problem statement and objectives, requirements, implementation, results, conclusion, and future scope.

3. LITERATURE SURVEY

Existing maximum-flow methods include Ford–Fulkerson and Edmonds–Karp. Ford–Fulkerson augments flow along available paths, while Edmonds–Karp uses BFS to select shortest augmenting paths. Residual graphs show remaining capacity and reverse edges that allow previous flow to be adjusted. This project presents these concepts in an interactive step-by-step visualization.

4. Problem Statement and Objectives

Problem Statement: A clear and specific articulation of the problem, outlining its importance.

To visualize the Edmonds–Karp algorithm and determine maximum flow using BFS-based augmenting paths.

Problem Description: A concise explanation of the issue or challenge that needs to be solved.

The project models a directed network with edge capacities, repeatedly finds augmenting paths using BFS, updates residual capacities, and computes the maximum flow from source to sink.

Objectives:

Identify the Project's Purpose:

Determine why the project was initiated and what business problem it aims to solve. Then frame the objectives of your project.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Computer or laptop with 4 GB RAM or higher
- Dual-core processor or equivalent
- Keyboard, mouse/touchpad, and display

Software Requirements:

- Operating System: Windows, Linux, or macOS
- Modern web browser
- HTML, CSS, and JavaScript
- Visual Studio Code or another code editor

6. IMPLEMENTATION

Modules: Flow Network Input; BFS Augmenting Path Search; Residual Capacity and Flow Update; Maximum Flow Result Visualization.

7. RESULTS & DISCUSSION

The application shows the flow network, BFS traversal, augmenting paths, residual capacities, and final maximum flow. Each step illustrates how the algorithm updates the network.

8. CONCLUSION & FUTURE SCOPE

The project visualizes Edmonds–Karp through BFS-based augmenting paths and residual-network updates. It demonstrates how repeated augmentation produces the maximum flow when no further augmenting path exists.

Future improvements may include editable networks, larger graphs, playback controls, exporting results, and comparison with other maximum-flow algorithms.

9. REFERENCES

1. Edmonds, J. and Karp, R. M., Theoretical Improvements in Algorithmic Efficiency for Network Flow Problems, 1972. https://doi.org/10.1137/0202019

2. CP-Algorithms, Edmonds–Karp Algorithm. https://cp-algorithms.com/graph/edmonds_karp.html

3. GeeksforGeeks, Edmonds–Karp Algorithm. https://www.geeksforgeeks.org/dsa/edmonds-karp-algorithm/

4. Wikipedia, Edmonds–Karp Algorithm. https://en.wikipedia.org/wiki/Edmonds%E2%80%93Karp_algorithm

5. Princeton University, Maximum Flow lecture notes. https://www.cs.princeton.edu/courses/archive/spr03/cs226/lectures/maxflow.pdf

6.

7.

8.

9.

10.